

★ TOPIC 00: ORIENTATION, PHILOSOPHY & MASTER ROADMAP

MASTER PAGE 1
OF 2

1 PURPOSE

Zero Fluff. Pure Senior/Staff FAANG Signal.

Engineered for Senior & Staff SWE offers at Google, Meta, Amazon, Uber, Datadog & Airbnb.

Outcome:
Instant pattern recognition + zero-bug Java templates.

2 TARGET AUDIENCE

 **Beginners:** Intuitive mental models without heavy math proofs.

 **FAANG Candidates:** Master NeetCode 150 & LC Hard patterns.

 **Staff Engineers:** 20-min pre-interview template cram sheet.

3 WHY THIS HANDBOOK IS DIFFERENT

University / Textbooks	This FAANG Handbook
Isolated proofs & history	Pattern Evolution & DNA
Generic pseudocode	Production-Grade Java 17+
Memorizing 1000 problems	29 Core Pattern DNA Cards
Ignores interviews	Verbatim Interview Scripts

💡 AHA MOMENT

Pattern DNA > Problem Count: Solving 1,000 random problems causes burnout. Understanding 29 Pattern DNA cards solves thousands automatically!

🏆 GOLDEN RULE

CLARIFY → THINK → CHOOSE DS → OPTIMIZE → CODE → EXPLAIN. Never start coding without stating your pattern!

👁️ JAVA ONLY

100% Java Collections API (ArrayDeque, PriorityQueue, HashMap). No generic pseudocode excuses.

⚡ SPEED DRILL

Write every template from memory in under 3 minutes. Zero syntax hesitation under pressure.

📖 COMPLETE 20-TOPIC SEQUENCED DEPENDENCY GRAPH (00 → 19)

PHASE 1: LINEAR & SEARCH

00 Orientation & Strategy

Start



01 Foundations (Big-O, Math, Bits)

Base



02 Arrays & Prefix Sum

Core



03 Strings & KMP / Hash

Core



04 Hashing & Frequency

Core



05 Two Pointers

Pattern



06 Sliding Window

Pattern



07 Binary Search & Bounds

Search

PHASE 2: STRUCTURES & TREES

08 Linked Lists (Fast/Slow)

Struct



09 Stack & Monotonic Stack

Mono



10 Queue & Monotonic Deque

Mono



11 Heap / PriorityQueue

Top-K



12 Trees (DFS / BFS / BST)

Heavy



13 Trie (Prefix Trees)

String

PHASE 3: GRAPHS, DP & META

14 Graphs (DFS, BFS, Topo, UF)

Heavy



15 Recursion & Backtracking

Search



16 Dynamic Programming

Heavy



17 Greedy Algorithms

Opt



18 Staff Data Structures (LRU/LFU)

Staff



19 Interview Mastery & Meta

Final

⚠️ **DEPENDENCY RULE:** Never jump to Graphs (14) or DP (16) without mastering Trees (12) and Recursion (15). Dynamic Programming is simply Recursion + State Caching!

THE UNIVERSAL 13-BLOCK CHAPTER ARCHITECTURE

1. RECOGNITION: Trigger keywords in problem statements mapping instantly to this pattern.

2. MENTAL MODEL: Intuitive visual/physical representation of how the algorithm operates.

3. VISUALIZATION: Step-by-step state diagrams, ASCII trees, and pointer movement.

4. CORE THEORY: Mathematical invariants and structural mechanics behind the pattern.

5. JAVA TEMPLATE: Production-grade Java code skeleton (cross-referenced to Appendix A).

6. LINE-BY-LINE: Detailed explanation of every variable, loop condition, and base case.

7. DRY RUN: Line-by-line variable state trace table on concrete representative inputs.

8. COMPLEXITY: Time & Space complexity analysis with mathematical justification.

9. EDGE CASES: Boundary conditions (null, empty arrays, duplicates, single elements).

10. COMMON MISTAKES: Top 3 failure points candidates make in live FAANG coding interviews.

11. INTERVIEW QUESTIONS: Senior-level follow-up questions asked by Google, Meta & Uber.

12. LEETCODE SET: Curated progression of representative problems (Easy → Medium → Hard).

13. REVISION SUMMARY SHEET: 1-Page condensed cheat card for 20-minute rapid pre-interview revision.

⚡ 30-DAY FAST TRACK

Experienced Revisers

- **Days 1-7:** Topics 01-07 (Arrays, Windows, BS)
- **Days 8-18:** Topics 08-14 (Trees, Tries, Graphs)
- **Days 19-25:** Topics 15-17 (Recursion, DP, Greedy)
- **Days 26-30:** Template Drills & Meta Mocks

🎯 60-DAY STANDARD

Recommended FAANG

- **Days 1-15:** Linear Structures & Search
- **Days 16-35:** Trees, Heaps & Graph Algorithms
- **Days 36-50:** Complete DP & Staff Structures
- **Days 51-60:** Book 2 Mocks & Pattern Cards

🏛️ 90-DAY MASTERY

Beginners / Staff L6

- **Month 1:** Foundations
- **Month 2:** Non-Linear Data
- **Month 3:** Advanced DP & Mocks



INTERVIEW WEEK & 3-HOUR CRAM STRATEGY

Timeframe	Action Plan	Target Resource
7 Days Before	Review all 29 Pattern DNA cards in Book 2 Section 2	Book 2 Section 2 (Pattern DNA)
3 Days Before	Write all 24 Master Java Templates from memory without checking code	Book 1 Appendix A (Templates)
1 Day Before	Read "If You See... Think..." Tables & Common Mistakes	Part 13 (Interview Meta)
3 Hours Before	Run 3-Hour Cram Sequence (Topic 00 Page 3)	Topic 00 Page 3 Table

✅ **READINESS CHECKLIST:** You have completed Topic 00! You understand the 20-Topic Dependency Graph, the 13-Block Chapter Architecture, and your personal Study Schedule. Turn the page to **Topic 01: Foundations (Big-O, Math & Bit Manipulation)**!